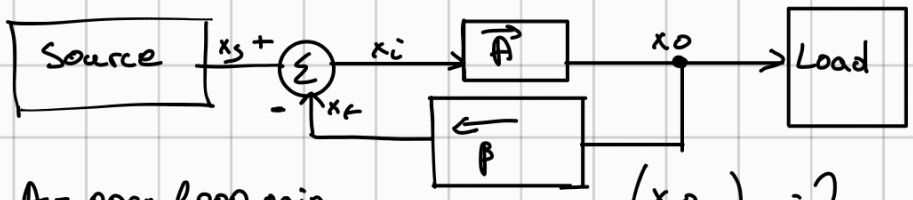
# Feedback

## The general feedback structure




A = open loop gain

β = feedback factor

$x_i$ = Input signal

$x_s$ = signal source

$x_o$ = output signal

$x_f$ = sampled output

$\left(\frac{x_o}{x_s}\right)_{CL} = ?$

$x_o = A x_i$

$x_f = \beta \cdot x_o$

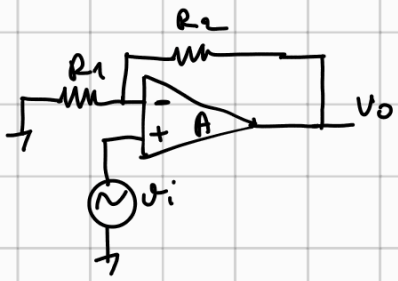
$x_i = x_s - x_f$

$x_i = x_s - \beta x_o$

$x_i = x_s - A\beta x_i$

$x_s = x_i (1 + A\beta)$

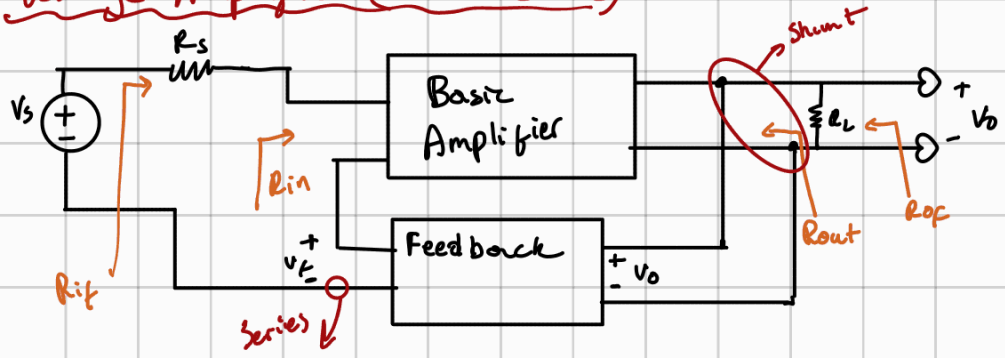
$\left(\frac{x_o}{x_s}\right) = \frac{A x_i}{x_i (1 + A\beta)} = \frac{A}{1 + A\beta}$



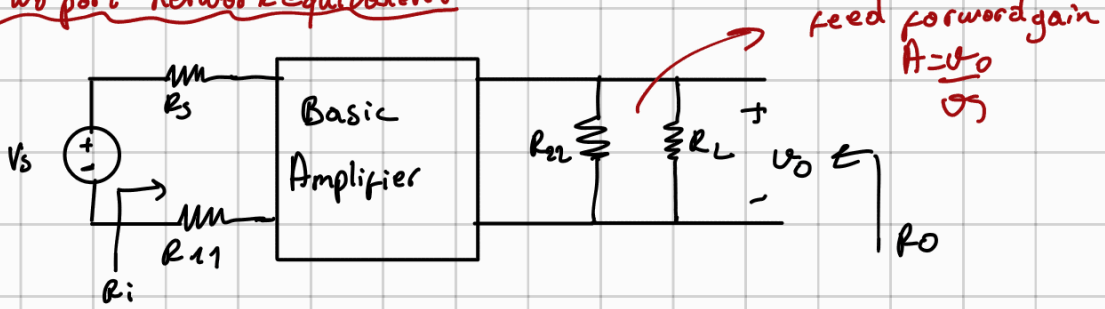

$\frac{v_o}{v_i} = \left(1 + \frac{R_2}{R_1}\right) \approx \frac{1}{\beta}$

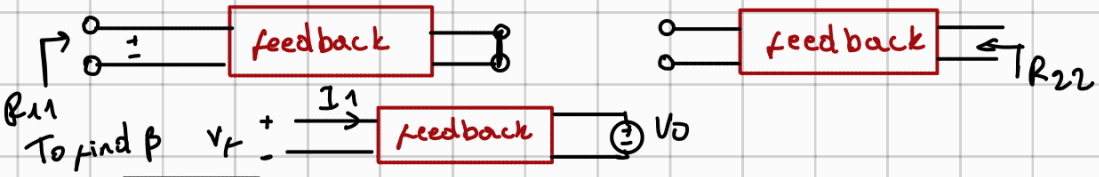
$\beta = \left(1 + \frac{R_2}{R_1}\right)^{-1}$

## Voltage Amplifiers (Shunt - Series)



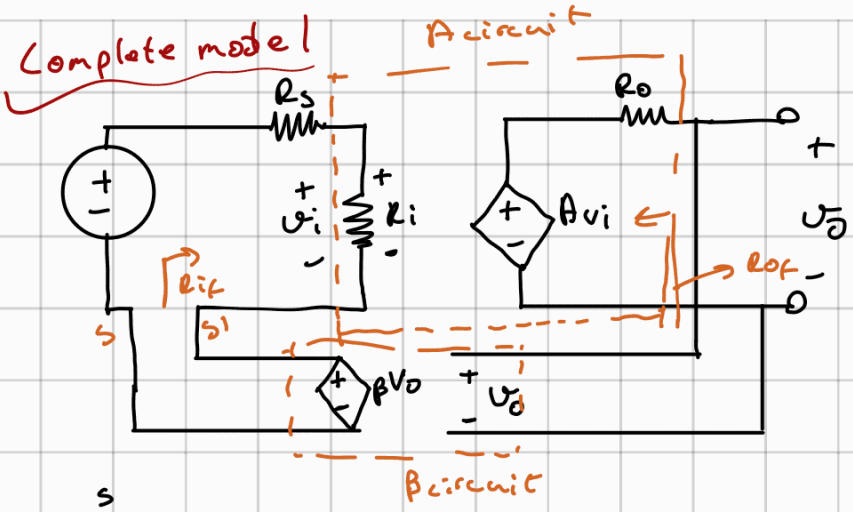

## Two port network equivalent

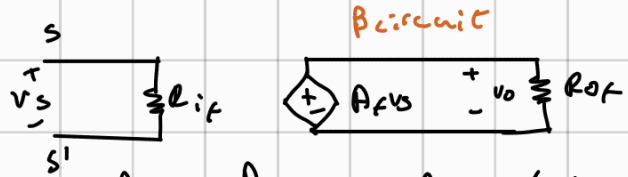



$\beta = \left.\frac{V_f}{V_o}\right|_{I_1}$

## Complete model

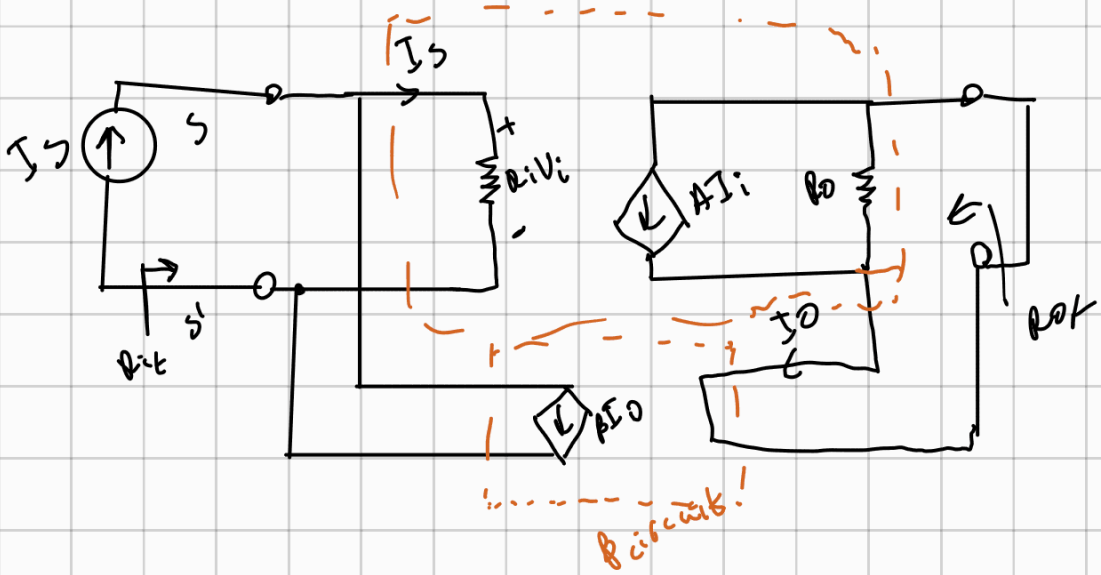


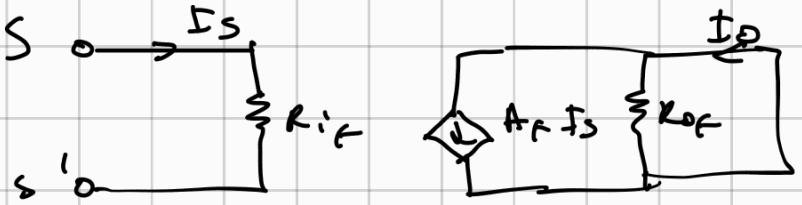




$$A_f = \frac{A}{1+A\beta} \qquad R_{if} = (1+A\beta) R_i$$

$$R_{of} = \frac{R_o}{(1+A\beta)}$$

## Lusren Amplifier feedback



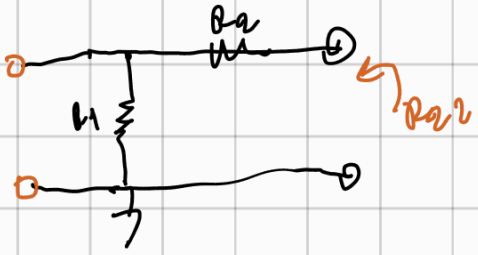

$$A_f = \frac{A}{1+A\beta}$$

$$R_{if} = R_i / 1+A\beta$$

$$R_{of} = R_o(1+A\beta)$$

Sorw



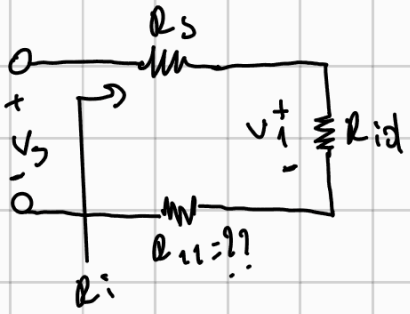

$R_{11} = R_1 // R_2$

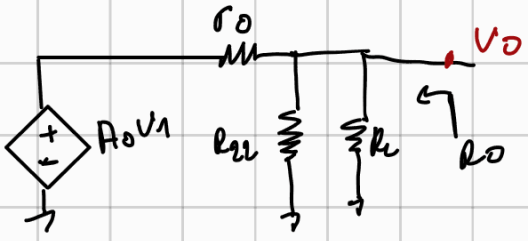
$R_{22} = R_1 + R_2$




$$\frac{V_f}{V_D} = \frac{R_1}{R_1+R_2} = \beta$$

$R_{eq} = R_S + R_{id} + R_{11} =$




$$\frac{v_0}{v_1} = \frac{R_{22} // R_L}{r_0 + R_{22} // R_L} \cdot A_0$$

$$A = \frac{v_0}{v_S} = \frac{v_0}{v_1} \cdot \frac{v_1}{v_S}$$

$$\frac{v_1}{v_S} = \frac{R_{id}}{R_S + R_{id} + R_{11}} \cdot \frac{v_S}{v_S}$$

$$A = \underbrace{\frac{R_{id}}{R_S + R_{id} + R_{11}}}_{R_i} \cdot \frac{R_{22} // R_L}{r_0 + R_{22} // R_L} \cdot A_0$$

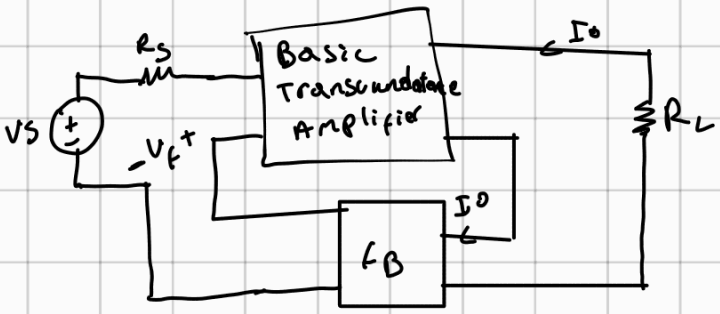
$$A_f = \frac{A}{1 + A\beta} \qquad \beta = \frac{R_1}{R_1 + R_2}$$

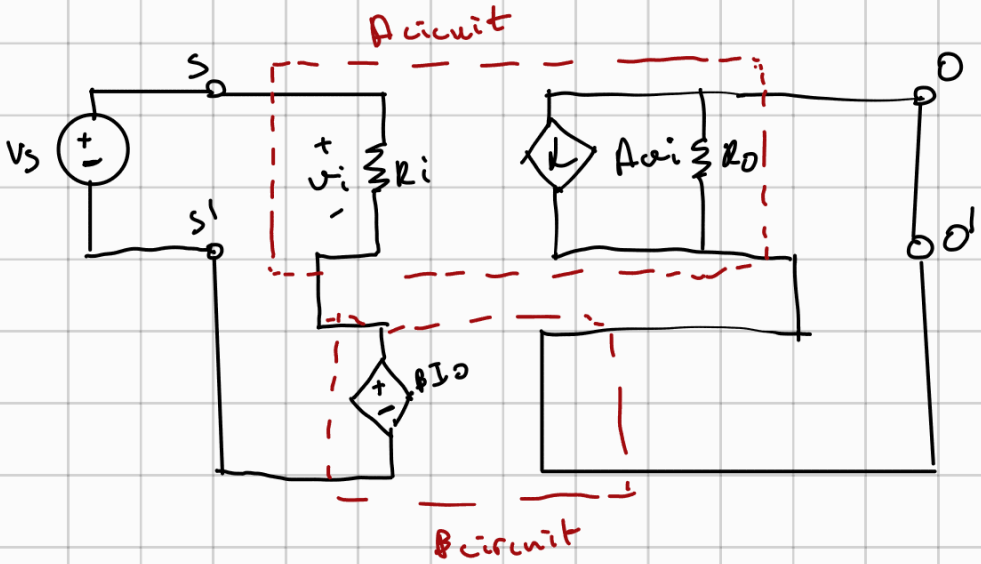
$R_{if} = (1+A\beta)R_i$ $\quad R_{of} = \frac{R_o}{(1+A\beta)}$

(V) (I)

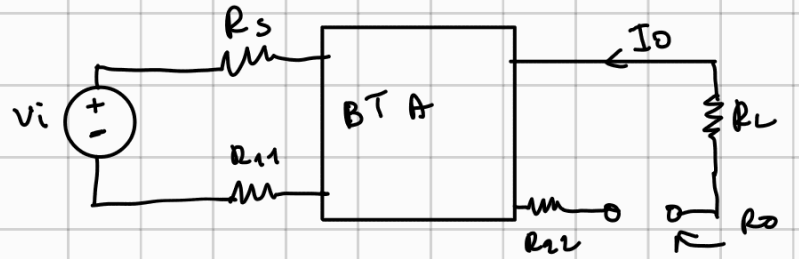
## Series - Series feedback Topology

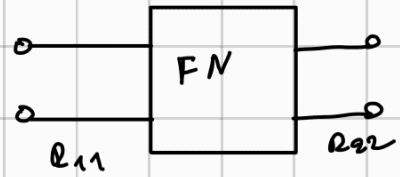
- Suitable for transconductance Amplifier



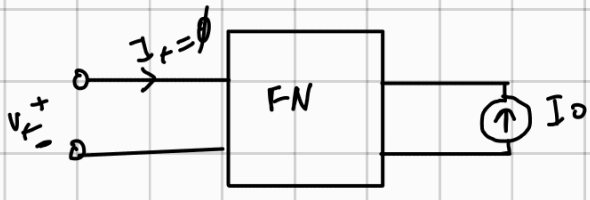




## Two part equivalent circuit




$\frac{I_o}{V_i} = A$



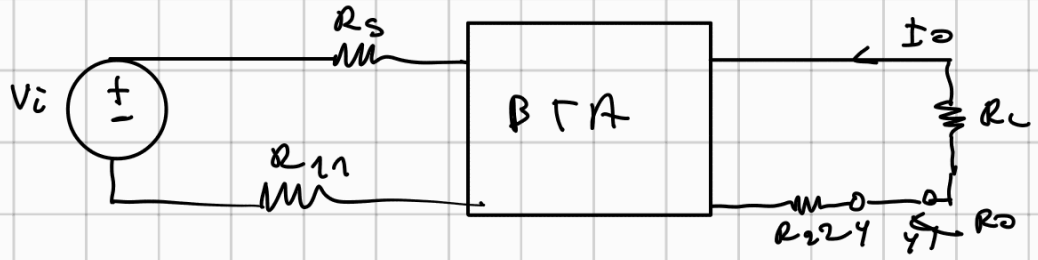

$\beta = \frac{V_f}{I_o}$

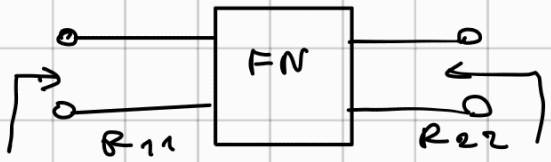
$R_{if} = R_i(1+A\beta)$

$R_{of} = R_o(1+A\beta)$

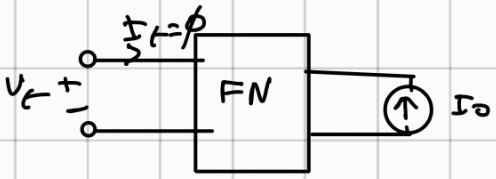
$A_f = \frac{A}{1+A\beta}$

# Two-port equivalent circuit




$$\frac{I_o}{V_1} = A$$



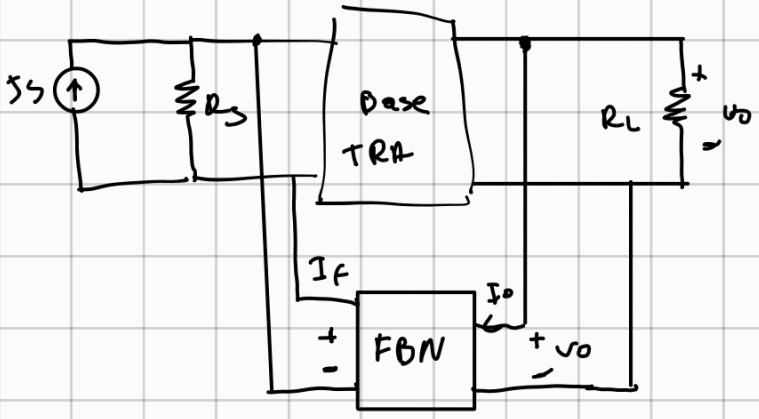

$$R_{if} = R_i(1+A\beta)$$

$$R_{of} = R_o(1+A\beta)$$

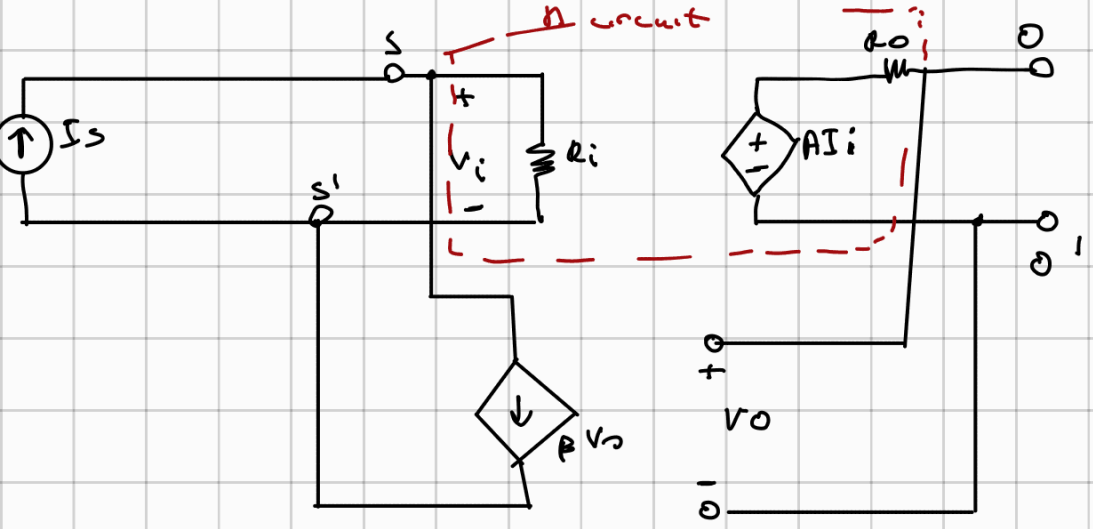
$$\beta = \frac{V_f}{I_o}$$

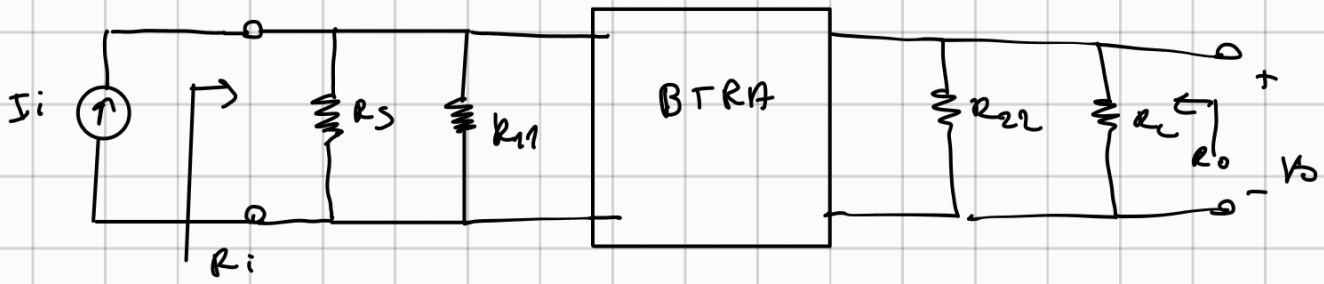
$$A_c = \frac{A}{1+A\beta}$$

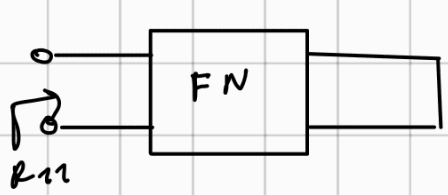
## Shunt-Shunt feedback topology
(Trans resistance amplifier)

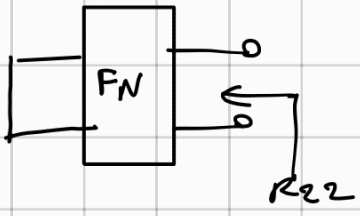


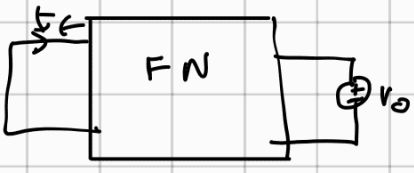

$$A = \frac{V_o}{I_s}$$





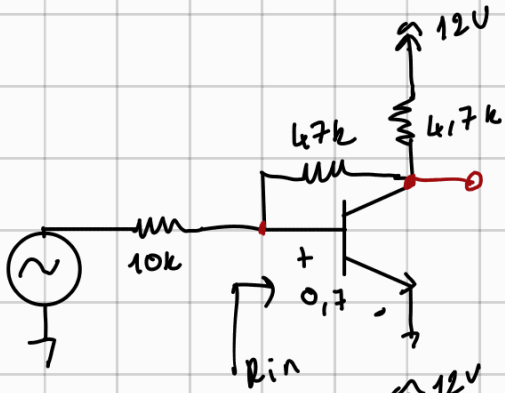

$\beta = \frac{I_f}{V_o}$

$R_{if} = \frac{R_i}{1+A\beta}$

$R_{of} = \frac{R_o}{1+A\beta}$

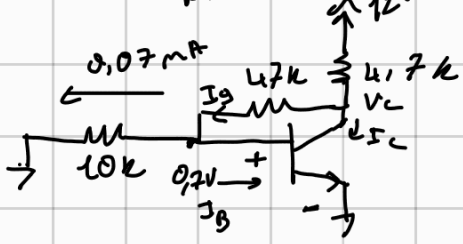
$A_f = \frac{A}{1+A\beta}$

(2)




ignore 60 || . $\beta = 100$
($V_A = \infty$)

DC




$\frac{0,7}{10k} = 0,07\ mA$

$\frac{12 - V_c}{4,7} = I_c + \frac{V_c - 0,7}{4\ 7k}$

$$I_C = \frac{12 - V_C}{4{,}7k} - \frac{V_C - 0{,}7}{4{,}7k}$$

$$I_C = \frac{120 - 11V_C + 0{,}7}{4{,}7k}$$

$$\frac{V_C - 0{,}7V}{4{,}7k} = I_B + 0{,}07mA$$

$$I_B = \frac{V_C - 0{,}7}{4{,}7k} - 0{,}07mA$$

$$\frac{100V_C - 70}{4{,}7k} - 7mA = \frac{120 - 11V_C + 0{,}7}{4{,}7k}$$

$$100V_C - 70 - 120 + 11V_C - 0{,}7 = 47.7$$

$$\frac{111V_C}{111} = \frac{47.7 + 120 + 70{,}7}{111}$$

$$V_C = 4{,}68\ V$$

$$I_C = 1{,}47mA$$

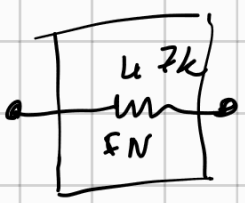
$$I_B = 0{,}0147mA$$

$$g_m = \frac{I_C}{V_t} = 58\ mS$$

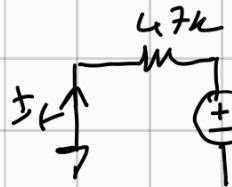
$$r_\pi = \frac{(\beta+1)}{g_m} = 1{,}7k\Omega$$

a) Feedback type
(Shunt-Shunt)

b) Feeback network

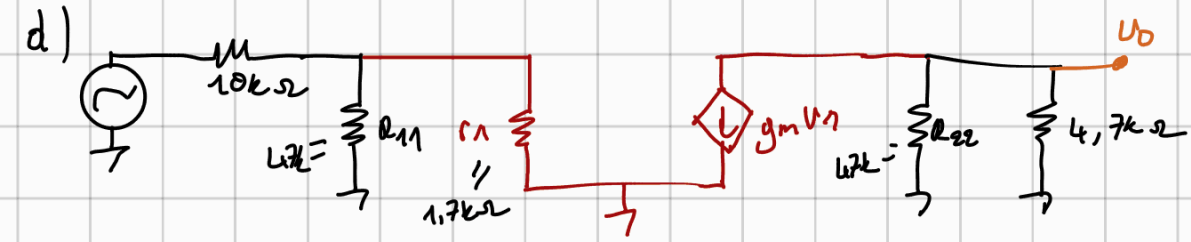


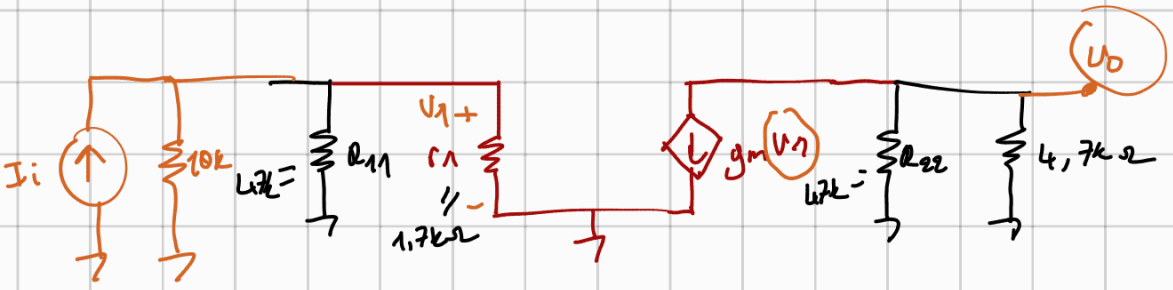

c) $R_{11} = R_{22} = 4{,}7k\Omega$




$$\frac{I_f}{V_o} = \frac{-1}{4{,}7k\Omega} = \beta$$

d)




$$A = \frac{V_o}{I_{in}} = \frac{v_o}{v_i} \cdot \frac{v_i}{I_{in}}$$

$$\frac{v_o}{v_i} = -(4{,}7k \,//\, 47k)\, g_m$$

$$\frac{v_i}{I_i} = \frac{I_i\,(10k \,//\, 47k \,//\, 1{,}7k)}{I_i}$$

$$A = -(4{,}7k \,//\, 47k) \cdot 58\,m \cdot \underbrace{(10k \,//\, 47k \,//\, 1{,}7k)}_{1{,}4k\Omega}$$

$$A = -341{,}04\ k\Omega$$

$R_i = 1{,}4k \quad R_o = 4{,}27k$

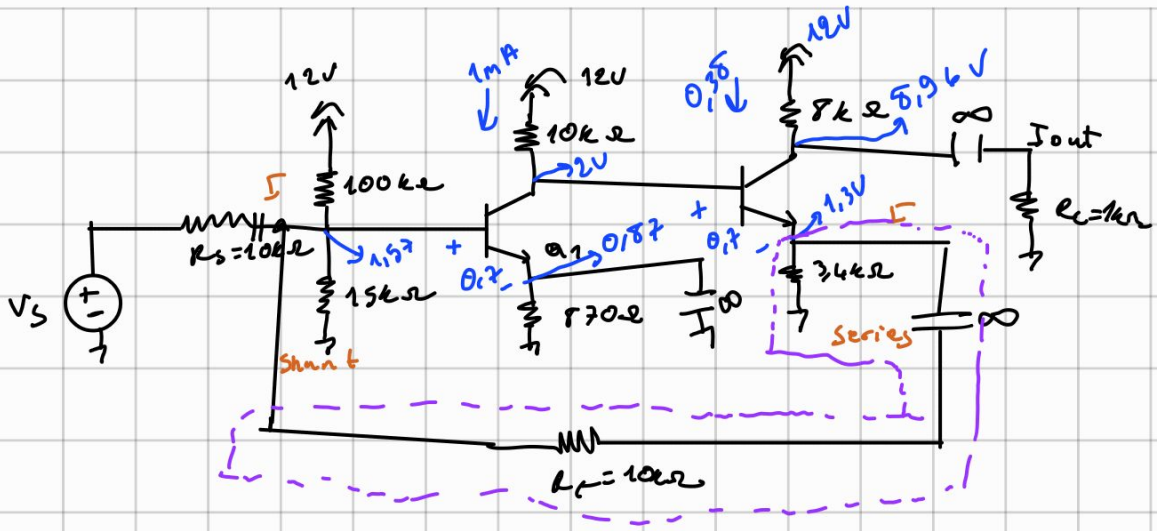
$$A_f = \frac{A}{1+A\beta} = -43{,}06k\Omega$$

$$R_{if} = \frac{R_i}{1+A\beta} \qquad R_{of} = \frac{R_o}{1+A\beta}$$

$$V_s = 10k\, I_i \qquad \frac{V_o}{V_s} = \frac{V_o}{10k \cdot I_i}$$

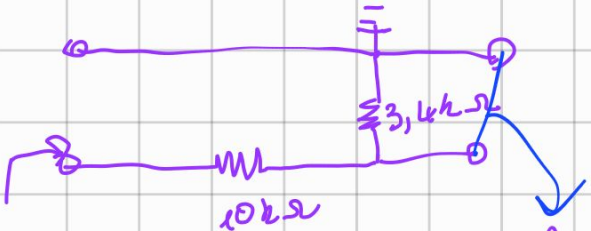
feedback

$$\frac{V_o}{I} = A_f$$




a) Series - Shunt

b)

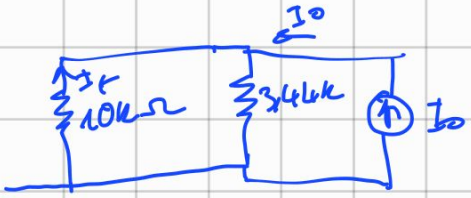



$R_{11} = 13{,}4\,k\Omega$

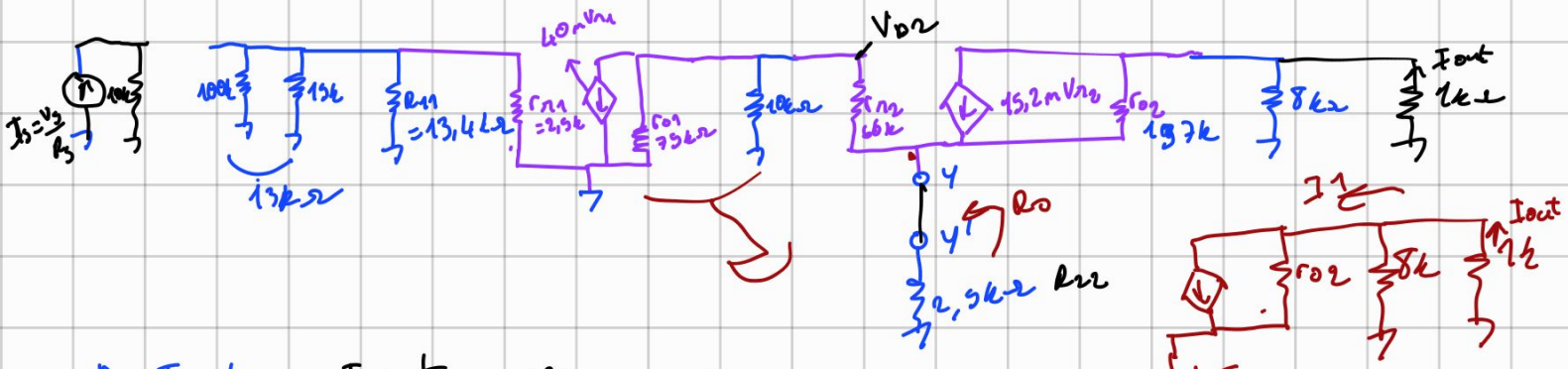
$R_{22} = 10k \,//\, 13{,}4k$

$\simeq 2{,}53k$

β




$\beta = \frac{I_f}{I_o} = -\frac{3{,}4k}{13{,}4k} \simeq -0{,}25\,[A/A]$




$A = \frac{I_{out}}{I_s} = \frac{I_{out}}{v_{n2}} \cdot \frac{v_{b2}}{v_{n1}} \cdot \frac{v_{n1}}{I_s}$

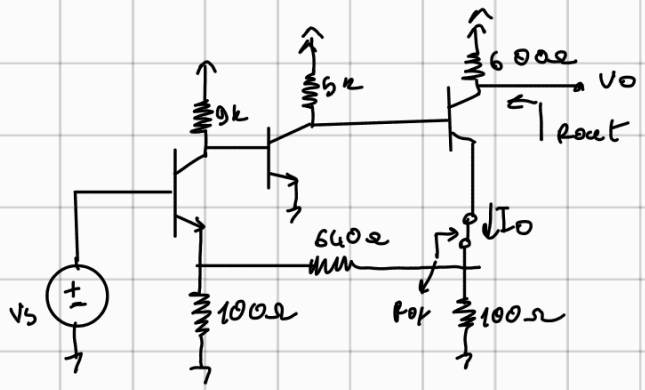
$V_{n1} = I_s (10k \,//\, 13{,}4k \,//\, 13k \,//\, 2{,}5k) = 1{,}55k\, I_s$

$V_{o2} = -g_{m1} V_{n1} (75k \,//\, 10k \,//\, (6{,}6k + 250k)) \simeq -920\, V_{n1}$

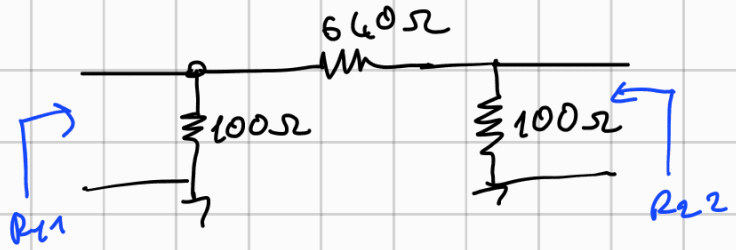
$(\beta+1)(2{,}5k)$

$$I_{out} = \frac{v_{bn}}{\frac{6,6k\Omega}{100} + 2,1k} \cdot \frac{7}{9}$$

$$\frac{I_{out}}{I_S} = -192,4\ [A/A]$$



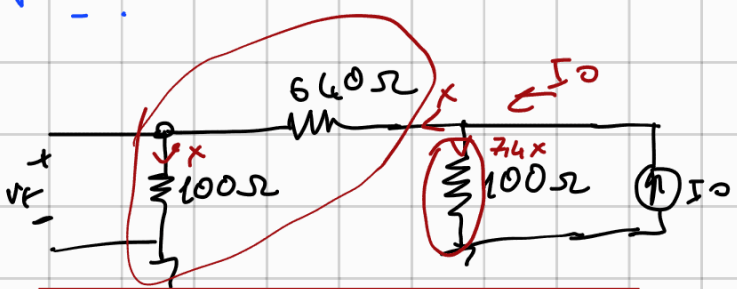

1) Series-Series Transconductance




$R_{11} = 100 // (640 + 100) = 88\ \Omega$

$R_{22} = 100 // (640 + 100) = 88\ \Omega$

B = ??




$$\frac{V_t}{I_O} = \frac{100x}{7,4x} = 11,9\ \Omega //$$

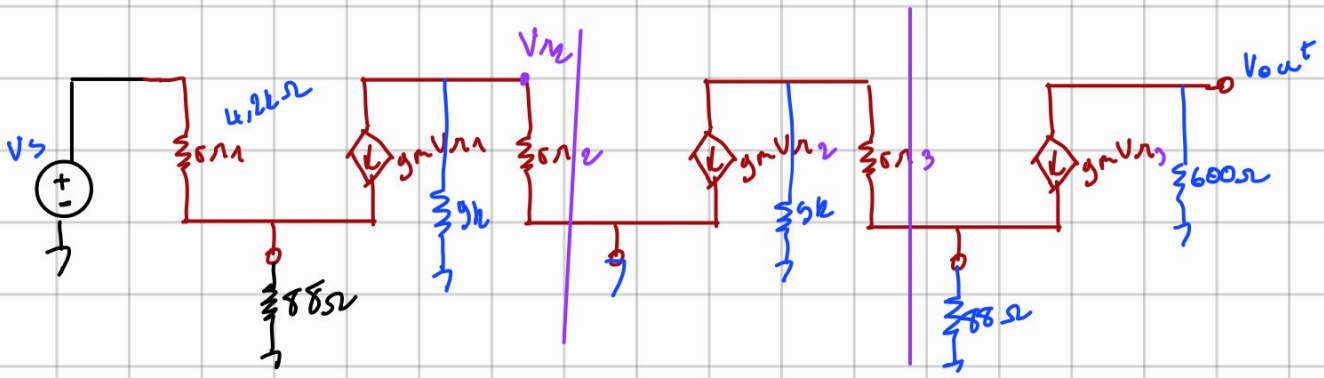
$$g_{m1} = \frac{I_c}{V_a} = 26mS$$

$g_{m2} = 40mS$

$g_{m3} = 10mS$

$r_{\pi_1} = \frac{(\beta+1)}{g_m} = 4,2k\Omega$

$r_{\pi_2} = 2,5k$ $r_{\pi_3} = 0,625k$




$$A = \frac{I_o}{V_s} = \frac{I_o}{v_{\pi 3}} \cdot \frac{v_{\pi 3}}{v_{\pi 2}} \cdot \frac{v_{\pi 2}}{v_{\pi 1}} \cdot \frac{v_{\pi 1}}{v_s}$$

$$I_o = g_{m3} V_{\pi 3} + \frac{V_{\pi 3}}{0{,}62k}$$

$$V_s = V_{\pi 1} + 88\left(0{,}024\, V_{\pi 1} + \frac{V_{\pi 1}}{4200}\right)$$

$$V_s \cong 3{,}03\, V_{\pi 1}$$

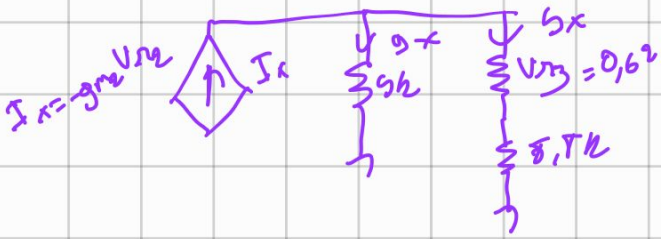
$$\frac{V_s}{V_{\pi 1}} = 3{,}03$$

$$V_{\pi 2} = -\frac{g_{m1} V_{\pi 1} \cdot 2{,}5k \cdot 9k}{11{,}5k} \quad (2{,}5k \text{ = } 4k \parallel 4k)$$

$$= 47 V_{\pi 1}$$

$$\frac{R_1 \| R_2}{R_1 + R_2}$$

$V_{\pi 3} =$



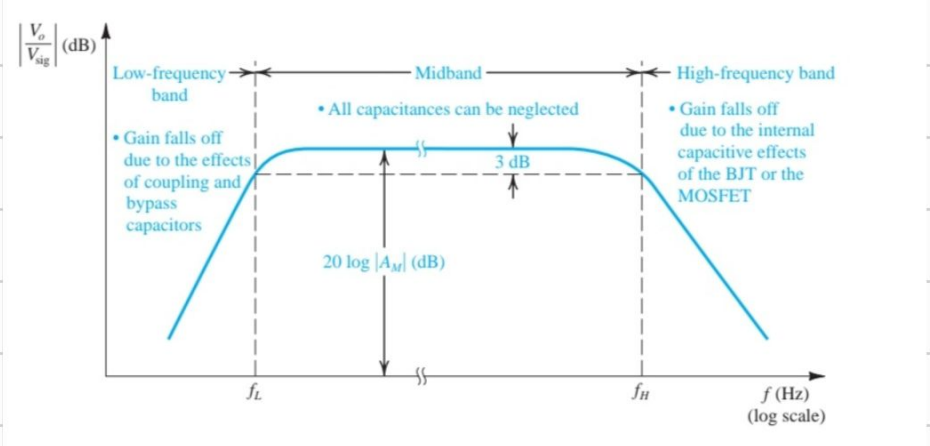

$$V_{\pi 3} = 3{,}1x = -8{,}86 V_{\pi 2}$$

$$I_x = 14x$$

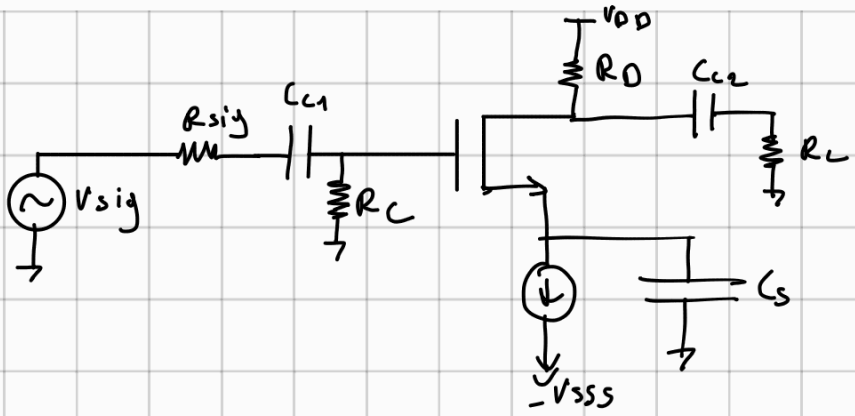
$$V_{\pi 3} = \frac{-g_m V_{\pi 2}}{14} \cdot 0{,}62k \; V_{\pi 2}$$

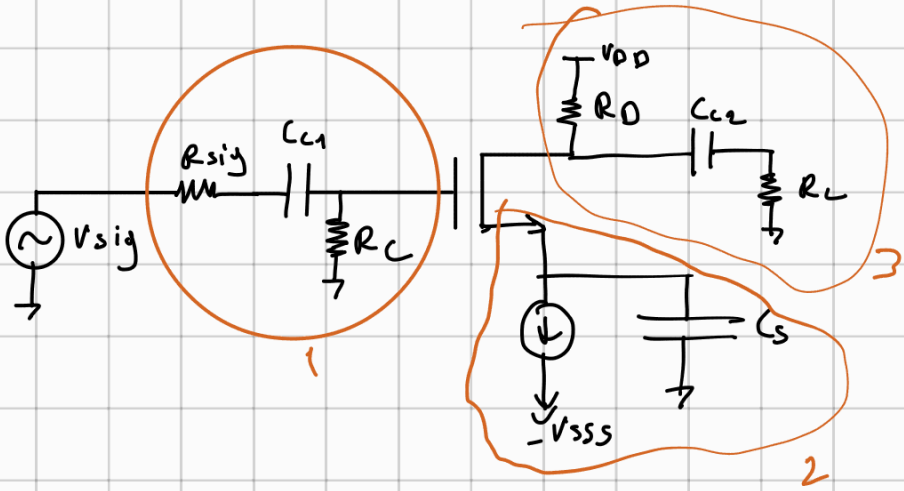
$$= -8{,}86\, V_{\pi 2}$$

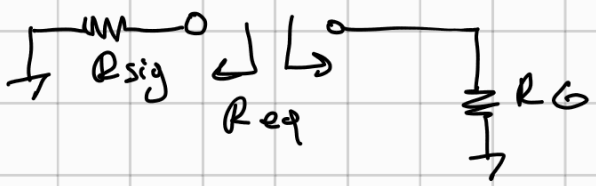
## Frequency Response



Low frekansta alt devrelerin analizi yapılırken C'ler short kabul edilir.



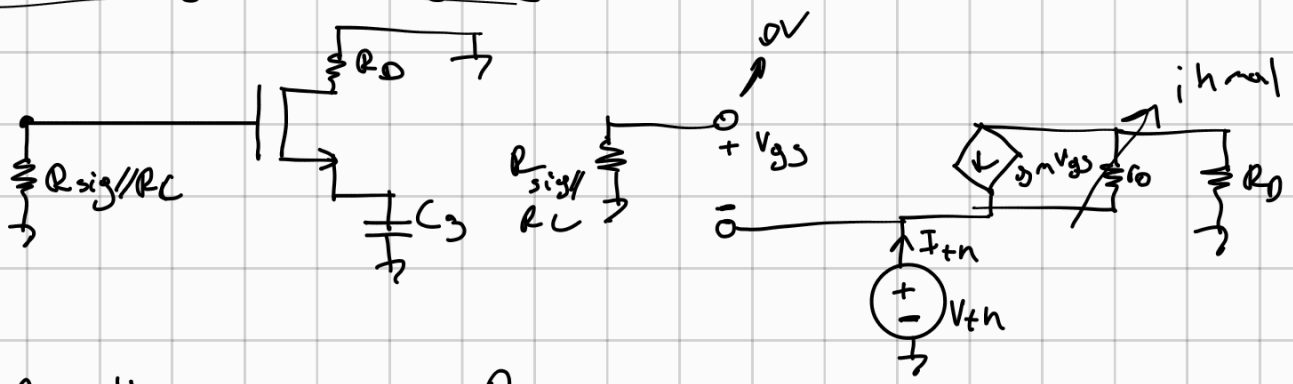




$$\omega_{p1} = \frac{1}{C_{C_1}(R_{sig}+R_E)}$$

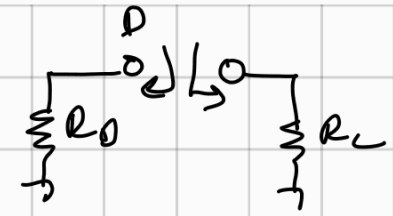
$$R_{eq} = R_{sig} + R_G$$

2. kapasiteden kaynaklı etkiyi bulalım




$$\frac{V_{tn}}{I_{tn}} = ? = \frac{-V_{gs}}{-g_m V_{gs}} = 1/g_m = R_{p2}$$

$$\omega_{p2} = \frac{1}{C_S(R_{p2})} = \frac{1}{\frac{C_S}{g_m}} = \frac{g_m}{C_S}$$




$$\omega_{p3} = \frac{1}{(R_D + R_L) C_{c2}}$$

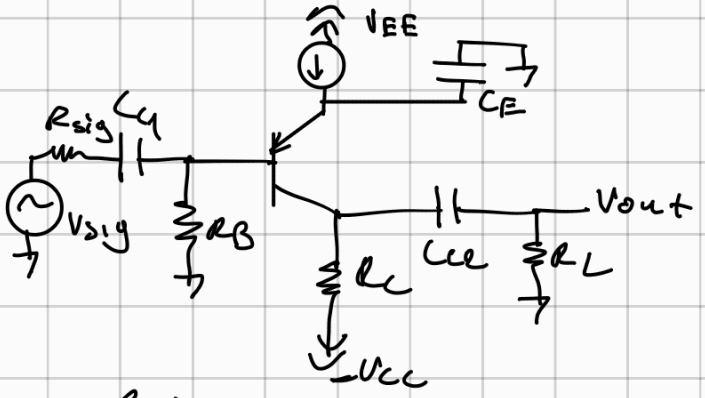
2 pole dominant olur çünkü direnci yüksek

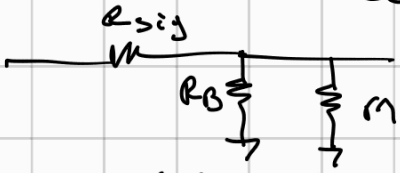
$$A_{mid} = -\left[\frac{R_G}{R_{sig} + R_G}\right] g_m (R_D // R_L)$$

$$A(s) = A_{mid}\left[\frac{s}{s+\omega_{p1}}\right]\left[\frac{s}{s+\omega_{p2}}\right]\left[\frac{s}{s+\omega_{p3}}\right]$$

$$f_c = \frac{1}{2\pi}\left[\frac{1}{R_1 C_1} + \frac{1}{R_2 C_2} + \frac{1}{R_3 C_3}\right]$$

Common emitter amplifier



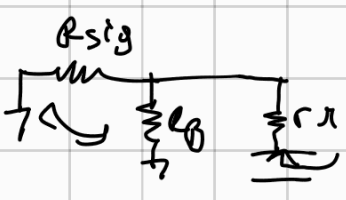




$$A_m = -g_m (R_C // R_L) \frac{(R_B // r_\pi)}{R_{sig} + (R_B // r_\pi)}$$

$$R_{p1} = (R_{sig} + R_B // r_\pi)$$

$$\omega_{p1} = \frac{1}{C_{C1}(R_{sig} + R_B // r_\pi)}$$

ce den bakarken

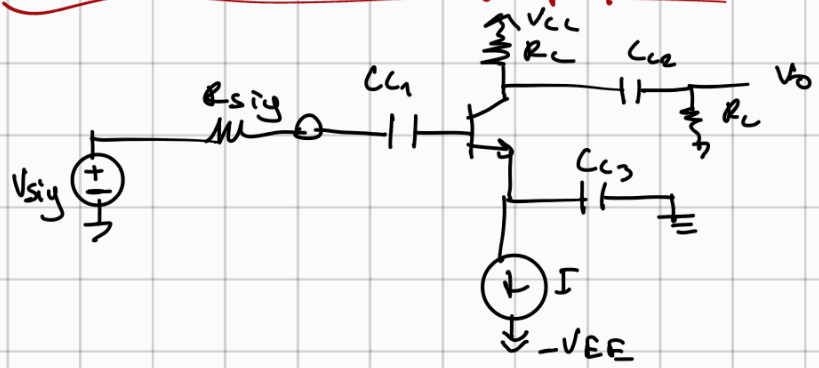


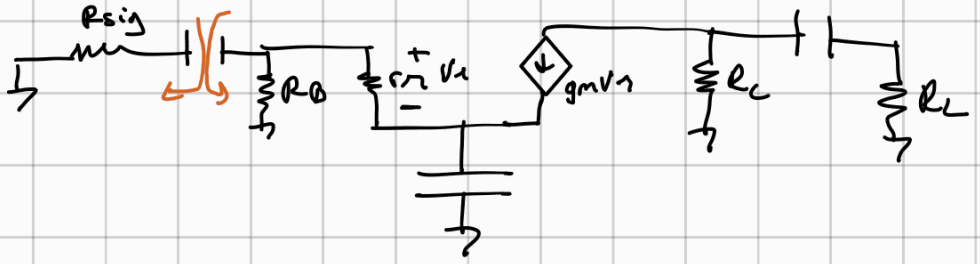
$$R_{p2} = R_{CE} = \frac{R_{sig} /\!/ R_B + r_\pi}{\beta + 1}$$

$$\omega_{p2} = \frac{1}{\frac{(R_{sig} /\!/ R_B) + r_\pi}{\beta + 1} C_E}$$

$$\omega_{p3} = \frac{1}{C_{c2}(R_C + R_L)}$$

## The Common Emitter Amplifier

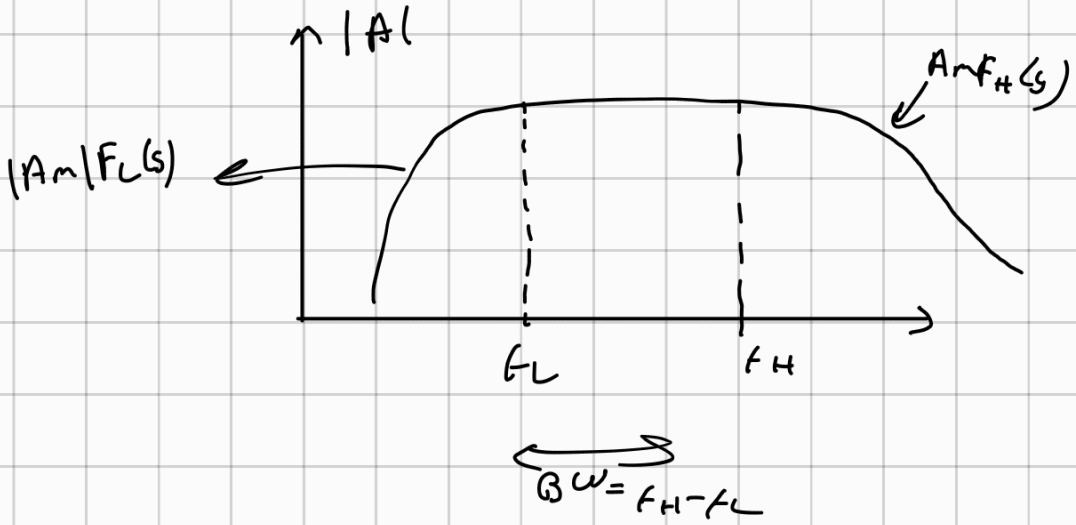


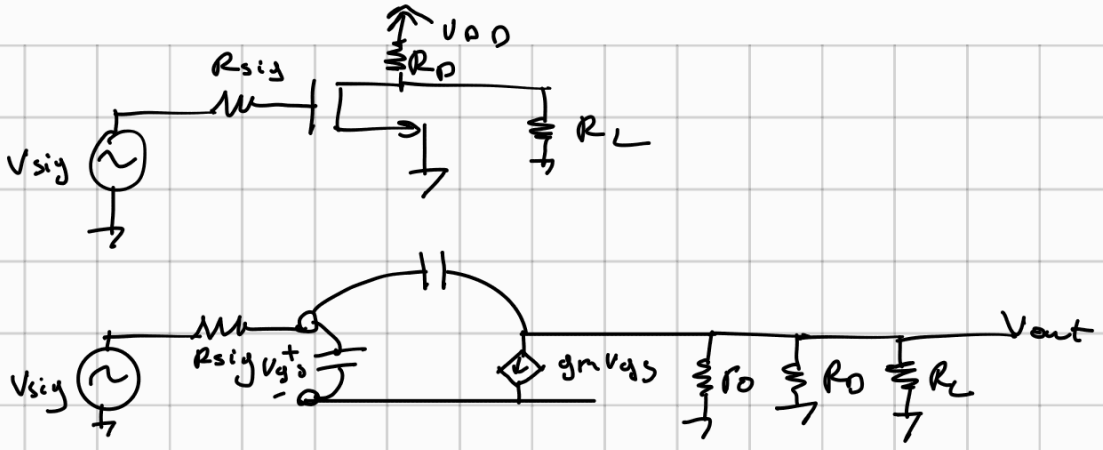




$$R_{cc1} = R_{sig} + R_B /\!/ r_\pi$$

$$R_{cc3} = \frac{R_{sig} /\!/ R_B + r_\pi}{\beta + 1} = r_e + \frac{R_{sig} /\!/ R_B}{\beta + 1}$$

$$R_{cc2} = R_C + R_L$$

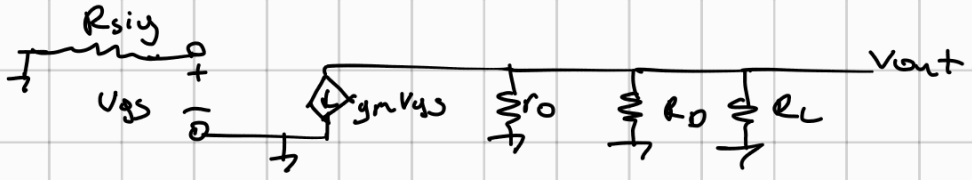




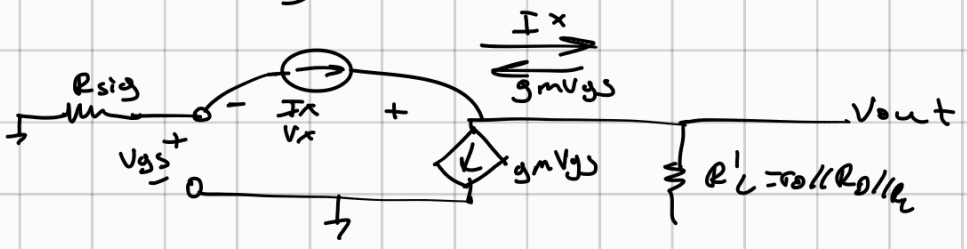




Analiz yaparken kısa devreye opsiyonlar




$R_{gs} = R_{sig}$




$\frac{V_x}{I_x} = ?$

$V_{out}: \quad V_x + V_{gs} = (I_x - g_m V_{gs}) R_L'$

$V_{gs} = -R_{sig} I_x$

$V_x - R_{sig} I_x = (I_x + g_m R_{sig} I_x) R_L'$

$V_x = I_x \left( R_{sig}(1 + g_m R_L') + R_L' \right)$

$\frac{V_x}{I_x} = R_{sig}(1 + g_m R_L') + R_L'$

$R_{gd} = R_{sig}(1 + g_m R_L') + R_L'$

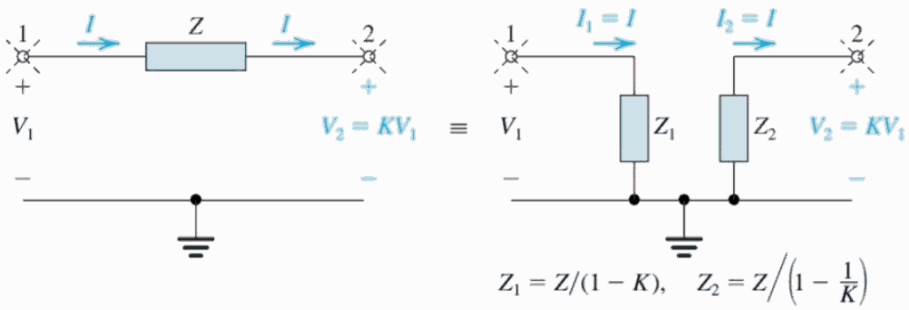
$R_{db} = R_L'$

$z = R_L$

$z_H = R_{gs} C_{gs} + R_{gd} C_{gd} + R_{db} C_{db}$

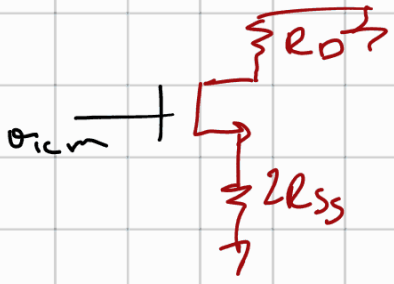
$f_H = \frac{1}{2\pi z_H}$

$$f_T = \frac{g_m}{2\pi(C_{gs}+C_{gd})}$$



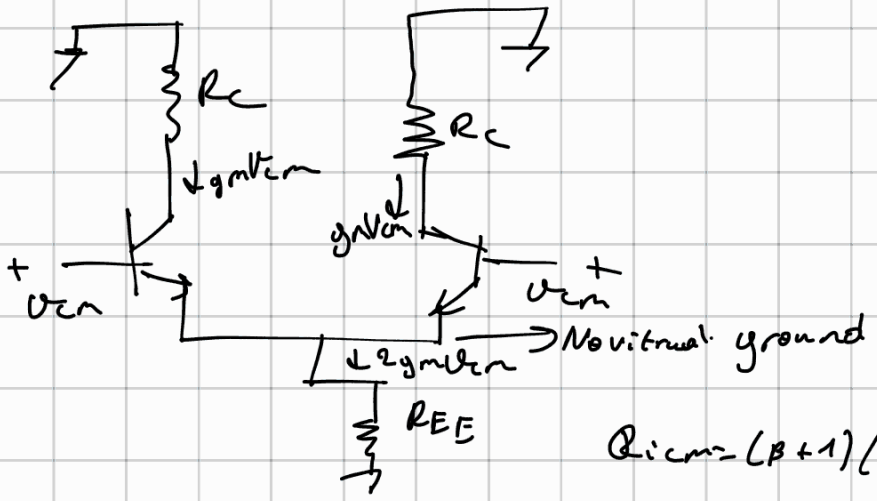

$$Z_1 = Z/(1-K), \quad Z_2 = Z\Big/\left(1-\frac{1}{K}\right)$$

Common mode half circuit



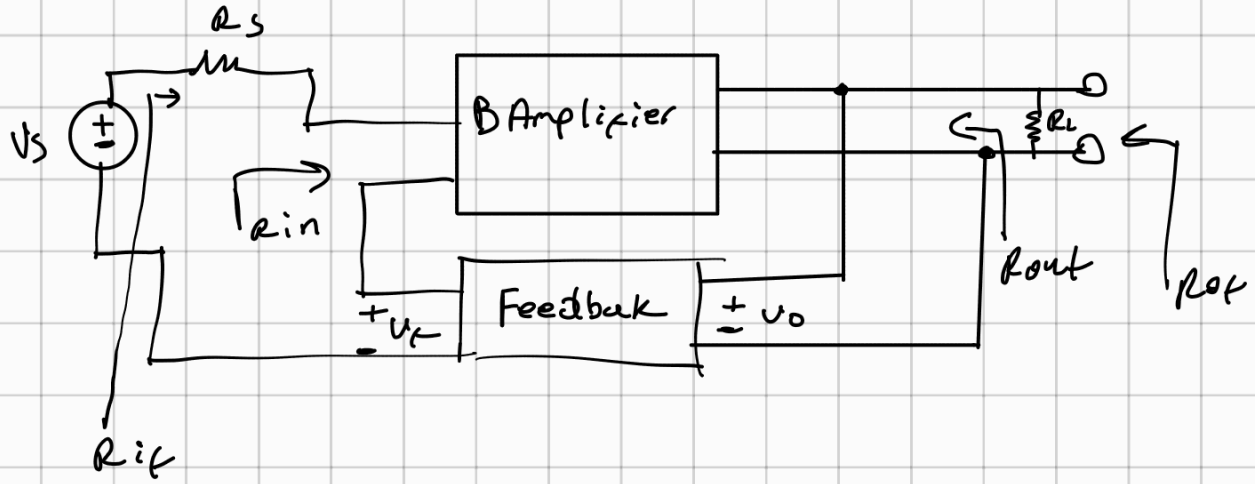

$$-\frac{R_D}{2R_{SS}}$$

$$CMRR_{single} = \frac{A_{dif\,single}}{A_{cm\,single}}$$




$$A_{icm} = (\beta+1)\left(R_{EE}+\frac{r_e}{2}\right)$$

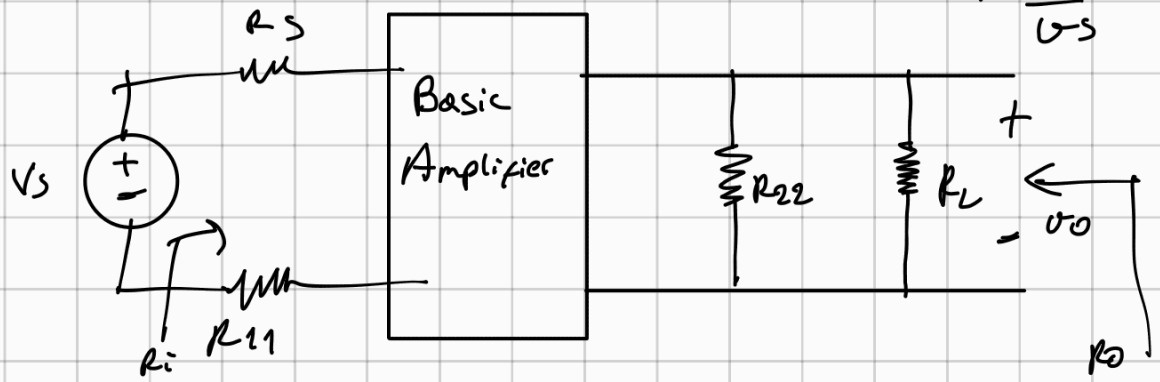
# 1- Voltage Amplifiers

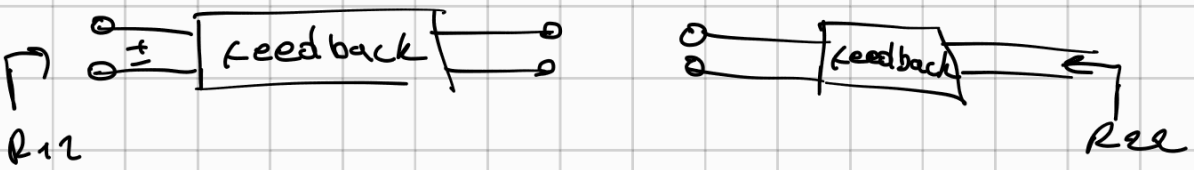



Two port network equivalent

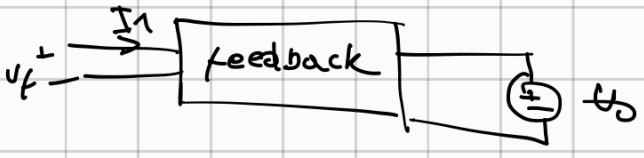
feedback voltage gain

$A = \frac{V_o}{V_s}$

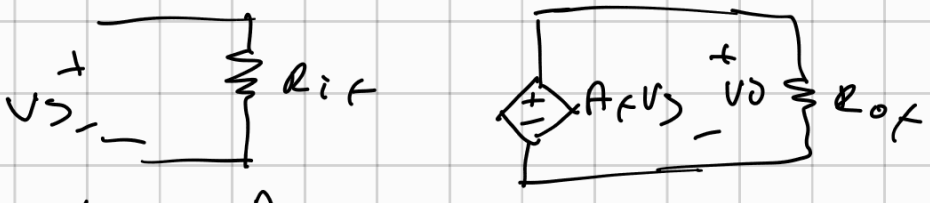



To find $\beta$




$$\beta = \frac{V_f}{V_o}\Big|_{I_1=0}$$

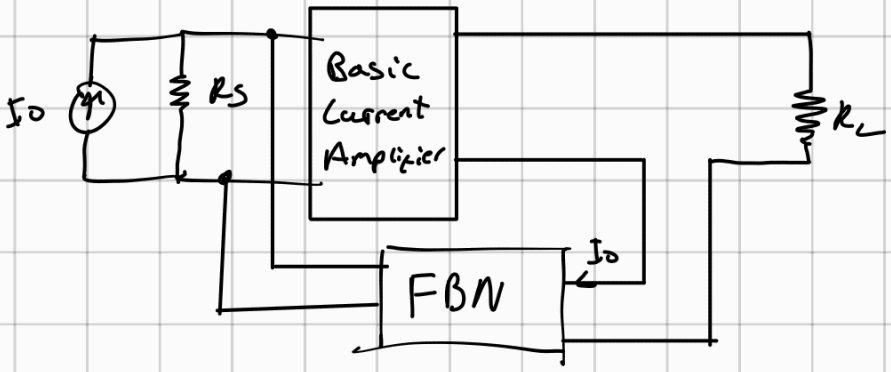



$$A_f = \frac{A}{1+A\beta}$$

$$R_{if} = (1+A\beta)R_i$$

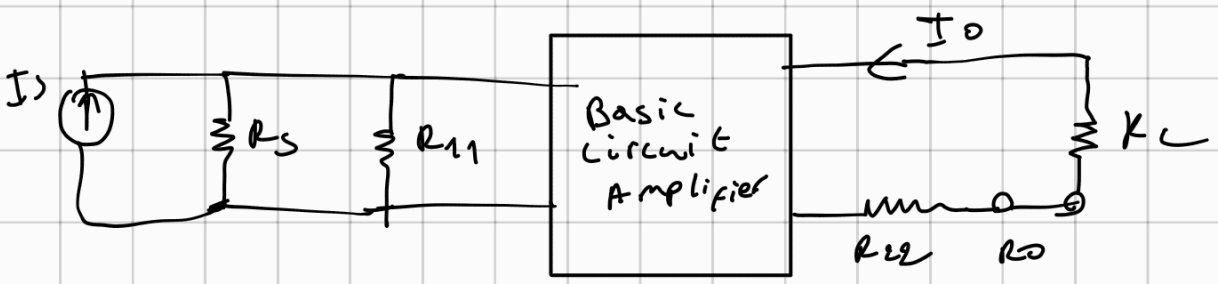
$$R_{of} = \frac{R_o}{1+A\beta}$$

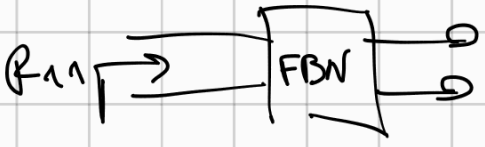
# 2- Current Amplifiers (Series-Shunt)

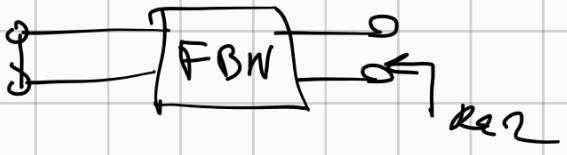



$$A = \frac{I_o}{I_i}$$

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$$\beta = \frac{I_f}{I_o}\Big|_{V_1=0}$$